

Elysium X 500 FR: Continued Training on Public Emotion Datasets for Multilingual Emotion Labeling over a 500-Coordinate Schema

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Abstract

We continue the training of Elysium X 500 FR, a LoRA adapter (rank 32, 36.9M trainable parameters) on Qwen2.5-1.5B-Instruct that returns a sparse JSON list of affective-state labels over a 500-coordinate schema in 12 languages. The original release was trained only on template-generated synthetic data. Here the same adapter is trained for one further epoch on those data plus 8,250 rows drawn from eleven public emotion datasets and mapped into the 500 coordinates with a hand-written label table. On the same frozen, sentence-disjoint internal test set of 1,195 rows, the released weights (run 1) reach exact label-set match of 23.33% when the English emotion name is absent from the sentence (403 rows; 19.35% before, +3.98 points) and 70.71% over all rows (69.54% before, +1.17 points). When the emotion name is present (792 rows) exact match is 94.82%, versus 95.08% before, a drop of two answers. We had set a bar of beating all three original numbers, so run 1 misses it, and we report that miss. A repeat run affected by a training-loop bug scored lower and is not the release. The test set is internal and synthetic, the mapping is our own, and we make no state-of-the-art claim.

1 Introduction

Elysium X 500 FR labels the emotion expressed by one named speaker at one turn, using a fixed vocabulary of 500 coordinates and a strength for each coordinate in 12 languages. The first technical report [1] trained it on 9,380 template-generated rows and reported the hard case first: with the English emotion name hidden from the sentence, exact match was 19.35% on 403 test rows. That report named the main weakness, which is that the training data are repetitive and mostly contain the label word, and listed adding more varied no-name data as the first fix.

This report tests that fix with free, public data. Public emotion corpora use their own label sets, so the central work is a mapping from each source's labels to the 500 coordinates. The model name and schema are unchanged. The weights are a new version.

Contributions. (1) A mapping table from 116 source labels in eleven public datasets to the 500-coordinate schema, released as a CSV. (2) A continued-training recipe that masks strength tokens on rows with no strength labels. (3) A re-measurement on the frozen internal test protocol of the first report, with the no-name, with-name and all-row subsets kept separate. (4) A disclosed account of a repeat run that failed because of a loop bug, and of a gate that the release does not fully pass.

2 Related Work

Emotion taxonomies and datasets. GoEmotions provides about 58k English Reddit comments labeled with 27 emotions plus neutral [4]. Ekman [7], Plutchik [8] and Cowen and Keltner [9] give small and mid-sized category sets. The 500-coordinate schema is a project design and is not claimed to be a validated psychological taxonomy. We know of no public evaluation on a 500-coordinate schema with the emotion word withheld, so we compare only with our own earlier release.

Efficient adaptation. We use LoRA [2] on Qwen2.5-1.5B-Instruct [3]. Training uses Transformers [5] and PEFT [6].

3 Task and Output Format

The input is a conversation of 1 to 7 turns up to and including a target turn, with the target speaker. Later turns are never given. The output is strict JSON with a **dimensions** list of objects, each holding an **id** and a **strength**, for example id 123 with strength 0.5. Here **id** is a coordinate from 1 to 500 and **strength** is one of 0.25, 0.5, 0.75 or 1.0. An omitted coordinate means unknown, not a checked negative.

4 Data

Team data. The team dataset is template-generated and synthetic, with wording monitored by a human reviewer (the author). “Human-monitored” means overseen and reviewed. It does not mean multi-annotator gold labels. We use 9,380 training rows, 1,188 development rows and 1,195 test rows, split by target text so that no test sentence appears in training. This is the same private test protocol as the first report. Of the 1,195 test rows, 792 are with-name (the English emotion name appears in the sentence) and 403 are no-name. The no-name rows cover 136 of the 500 coordinates.

Public data and mapping. Table 1 lists the sources. Each has its own labels, for example “admiration” in GoEmotions. The mapping table assigns each of 116 source labels to one or more of the 500 coordinates: 95 are exact matches, 17 are closest matches, 2 neutral labels are dropped and 2 labels with no match are dropped. Rows whose labels cannot be mapped are left

out. Closest matches are judgment calls, for example anticipation and pessimism in SemEval-2018. The mapping was written by us and has not been reviewed by the dataset authors. In total 8,250 public rows are used.

Source	Licence as listed by the source	Rows
GoEmotions	Apache-2.0	1,500
BRIGHTER, categories	CC BY 4.0	1,100
BRIGHTER, intensities	CC BY 4.0	1,100
EmpatheticDialogues	CC BY-NC 4.0	900
XED	CC BY 4.0	800
SemEval-2018 Task 1 E-c	unknown	700
daily_dialog	CC BY-NC-SA 4.0	550
ISEAR	not stated	500
dair-ai/emotion	“other”	450
tweet_eval (emotion)	unknown	350
Chinese Multi-Emotion Dialogue	MIT	300
Team dataset (train part)	CC BY-NC-SA 4.0	9,380

Table 1: Training sources. MELD (GPL-3.0), gated datasets and image-caption datasets were skipped.

Because some sources have an unknown or unstated licence and some are non-commercial, the weights are released for non-commercial research under CC BY-NC-SA 4.0. We do not call the release open source in the strict sense. Any dataset owner can ask for removal.

5 Method

Base and adapter. Qwen2.5-1.5B-Instruct with a rank-32 LoRA adapter (alpha 64, dropout 0.05). Training continues from the final adapter of the original release, not from the base model.

Strength masking. The team labels carry a strength and the exact-match metric includes it. Public rows have no strengths, so the strength tokens of public rows are removed from the training loss. The exception is the BRIGHTER intensity data, whose intensities are used.

Training. One epoch over 17,630 rows (1,102 optimizer steps), batch size 4 with 4-step gradient accumulation, learning rate 1×10^{-4} with cosine decay, maximum length 1,024 tokens, fp16, seed 501. Loss fell from 0.286 at step 20 to about 0.10. Model choices were made on the development split only. The test split was scored once per run. A single seed was used for the released run.

6 Evaluation Protocol

The protocol is that of the first report. Invalid JSON counts as a failure. We report exact match (the predicted coordinate-to-strength dictionary equals the gold one), top-1 (the first predicted coordinate is in the gold set), id-F1 (per row $2|G \cap P|/(|G| + |P|)$ over coordinate IDs, strengths ignored, averaged over rows) and micro-F1 (true positives, false positives and false negatives

summed over rows). All results are agreement with the template-generated, human-monitored labels, not with independent annotators.

7 Results

7.1 Headline: no-name rows

Table 2 gives the test results. On the 403 no-name rows run 1 reaches 23.33% exact match, up from 19.35%. Roughly three of four such rows are still wrong. Over all rows exact match is 70.71% (+1.17 points). On the 792 with-name rows exact match is 94.82%, which is 0.26 points (two answers, 751 versus 753) below the original.

Test, exact match (%)	n	Original	Run 1 (released)	Run 2
No-name (headline)	403	19.35	23.33	22.58
With-name	792	95.08	94.82	93.94
All rows	1,195	69.54	70.71	69.87
Micro-F1, all rows	1,195	70.27	71.78	71.76
Micro-F1, no-name rows	403	20.47	24.24	n/r

Table 2: Frozen internal test set (sentence-disjoint split). Run 1 and run 2 produced valid JSON on all 1,195 rows. n/r: not recorded.

On the development set (300 rows) run 1 reaches exact match 66.0% (no-name 14.3%, with-name 96.8%) and run 2 reaches 63.7% (no-name 13.4%, with-name 93.6%).

7.2 Gate

Before training we set a bar of beating all three original numbers. Run 1 beats two (no-name and all rows) and misses the with-name number by 0.26 points. The weights are released anyway on the decision of the project owner, and the miss is stated here and on the model card.

7.3 Per language, no-name rows

Run 1 no-name exact match per language: German 50.0, Portuguese 45.5, English 39.5, French 35.7, Korean 27.8, Hinglish 25.0, Mandarin 23.1, Arabic 21.1, Japanese 17.2, Spanish 16.4, Bengali 7.1, Hindi 3.8 (percent). Each language has few no-name rows, so differences of several points are within noise. Hindi rows are mostly romanized text.

7.4 The repeat run

We planned a two-epoch repeat (run 2, seed 502). The data loop ran once, so the run trained for one epoch (1,100 steps) while the learning-rate schedule was sized for 2,204 steps, and the rate never fell to zero. Run 2 scored lower than run 1 on all three checks. We did not run a

third time, because scoring the same test set again would weaken the claim that it is used once. Run 2 is reported for transparency and is not the release.

8 Limitations

- **Low no-name accuracy.** 23.33% means about three of four hidden-word sentences are wrong.
- **Template-generated team data.** Of 12,000 target lines in the team workbook, 8,936 are unique, and the written quality-review sheet covers about 100 rows, so labels are human-monitored, not gold.
- **Mapping by our rules.** A mapped label approximates the source label. The mapping has not been reviewed by dataset authors.
- **Unknown source licences.** SemEval-2018, ISEAR, dair-ai/emotion and tweet_eval carry unknown or unstated licences, which is why the release is non-commercial.
- **Possible translation leakage and narrow coverage.** The split is sentence-disjoint but translations may cross splits, and the no-name test rows cover 136 of 500 coordinates.
- **One run per setting, one internal test set.** There are no confidence intervals. A 1.17-point gain on 1,195 rows is small, and the 3.98-point no-name gain on 403 rows comes from a single run.

9 Future Work

Re-measure on an independently annotated, naturalistic no-name set. Review the mapping with the dataset authors or with several annotators. Group translations of the same sentence into one split. Add Devanagari Hindi data. Repeat the second epoch with the fixed loop and report both runs against a fresh held-out set.

10 Ethics and Intended Use

The model labels expressed text for one speaker at one turn. It cannot see private feelings, and it cannot establish diagnosis, intent, truthfulness or consent. It must not be used for clinical, diagnostic or crisis decisions, for surveillance or profiling of individuals, or to write memories or make commitments about a person without separate safeguards. On the no-name task its outputs are wrong most of the time. The team data are synthetic; the public datasets were used under the licences their pages list. The adapter inherits the limits and licence terms of the Qwen2.5-1.5B-Instruct base (Apache 2.0).

11 Conclusion

Continued training on mapped public data raises the no-name exact match of Elysium X 500 FR from 19.35% to 23.33% and the all-row score from 69.54% to 70.71%, and lowers the with-name score by two answers (95.08% to 94.82%). The release misses its own three-part bar by that margin, and a repeat run did worse because of a loop bug. The result is a small, honest improvement on a hard internal test, not evidence of general emotion understanding.

Availability

Hugging Face (open-nhe organization): model Elysium-X-500-FR, GGUF Q8_0 build Elysium-X-500-FR-GGUF, and dataset elysium-x-500-fr-dataset with the mapping table. Code and a plain-language guide: github.com/itsppm76/Elysium-X-500-FR. This report: doi:10.5281/zenodo.23179294. The weights and mapping table are released under CC BY-NC-SA 4.0 terms and this report record under CC BY-NC-ND 4.0; third-party datasets keep their own licences.

References

- [1] P. P. Mohanty. Elysium X 500 FR: A First Public Baseline for Multilingual Emotion Labeling over a 500-Coordinate Schema When the Emotion Word Is Hidden. Zenodo, doi:10.5281/zenodo.23159402, 2026.
- [2] E. J. Hu, Y. Shen, P. Wallis, Z. Allen-Zhu, Y. Li, S. Wang, L. Wang, W. Chen. LoRA: Low-Rank Adaptation of Large Language Models. arXiv:2106.09685, 2021.
- [3] Qwen Team. Qwen2.5 Technical Report. arXiv:2412.15115, 2024.
- [4] D. Demszky, D. Movshovitz-Attias, J. Ko, A. Cowen, G. Nemade, S. Ravi. GoEmotions: A Dataset of Fine-Grained Emotions. In Proc. ACL, 2020. arXiv:2005.00547.
- [5] T. Wolf et al. Transformers: State-of-the-Art Natural Language Processing. In Proc. EMNLP: System Demonstrations, 2020.
- [6] S. Mangrulkar, S. Gugger, L. Debut, Y. Belkada, S. Paul, B. Bossan. PEFT: State-of-the-art Parameter-Efficient Fine-Tuning methods. <https://github.com/huggingface/peft>, 2022.
- [7] P. Ekman. An argument for basic emotions. *Cognition and Emotion*, 6(3–4):169–200, 1992.
- [8] R. Plutchik. A general psychoevolutionary theory of emotion. In *Emotion: Theory, Research, and Experience*, Vol. 1, pp. 3–33. Academic Press, 1980.
- [9] A. S. Cowen, D. Keltner. Self-report captures 27 distinct categories of emotion bridged by continuous gradients. *PNAS*, 114(38):E7900–E7909, 2017.